KP15 Polynomial Classifiers

> Level: undergraduate, year 3
>
> Pattern Recognition | Chapter 4, §4.14

Slides: read and download

Learning objectives

- Write a second-order polynomial discriminant as a linear classifier on new coordinates and solve XOR with one interaction term.
- Count the terms with $k = (l + r)!/(r!\, l!)$ and explain the combinatorial explosion.
- Explain the power and the price of high-order polynomials (memory, overfitting, numerical instability) and why a kernel inner product avoids the explosion.

KP15 Polynomial Classifiers

One-sentence summary: A polynomial of order at most r as the discriminant can be as accurate as you like in theory but the number of terms explodes in practice; the power comes from the interaction terms and so does the disaster.

1. Order-2 expansion

Plain explanation: KP14 said lifting raises the probability of separable labels. The most direct way to build the new coordinates writes out every product.

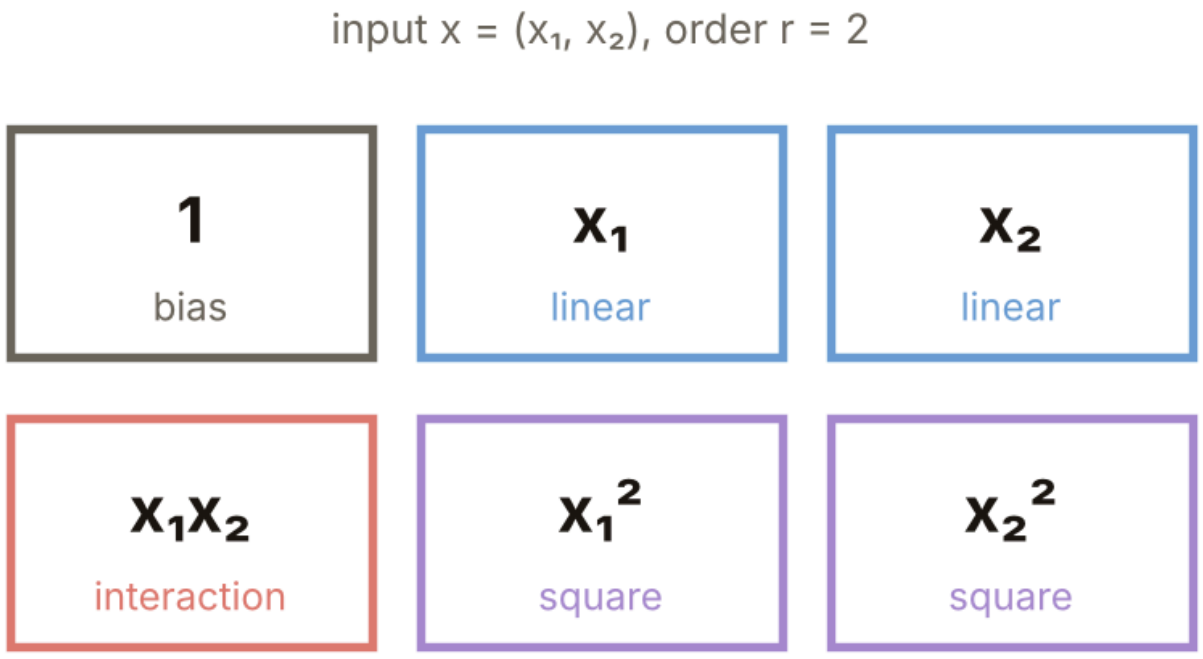
- **Polynomial classifier**: a polynomial of order at most r as the discriminant function, i.e. every product with total power at most r becomes a new coordinate before learning a linear classifier.
- **Interaction term**: a product of two or more different inputs such as $x_1 x_2$; a lone x_1^2 is a square term.

Formal definition (two inputs, order 2):

$$g(x) = w_0 + w_1 x_1 + w_2 x_2 + w_{12} x_1 x_2 + w_{11} x_1^2 + w_{22} x_2^2,$$

with new coordinates $y = [x_1, x_2, x_1 x_2, x_1^2, x_2^2]$ and the bias making 6 terms; $g = w^T y + w_0$ is linear in the 6 weights.

Figure `lec05-kp15-terms`: the six boxes of the order-2 expansion.



2. XOR lacks only one interaction term

Take only $y = (x_1, x_2, x_1 x_2)$: $(0,0) \to (0,0,0)$, $(0,1) \to (0,1,0)$, $(1,0) \to (1,0,0)$, $(1,1) \to (1,1,1)$. The plane $y_1 + y_2 - 2y_3 - \frac{1}{4} = 0$ gives back $g = -\frac{1}{4} + x_1 + x_2 - 2x_1 x_2$ and the four values $-0.25, +0.75, +0.75, -0.25$ all have the right sign. One interaction term $x_1 x_2$ solves XOR (the constant may be $-\frac{1}{4}$ or the -0.5 of KP13).

Figure `lec05-kp15-xor`: new coordinates, g and class of the four points.

x_1	x_2	$y_3 = x_1x_2$	g	class
0	0	0	−0.25	B
0	1	0	+0.75	A
1	0	0	+0.75	A
1	1	1	−0.25	B

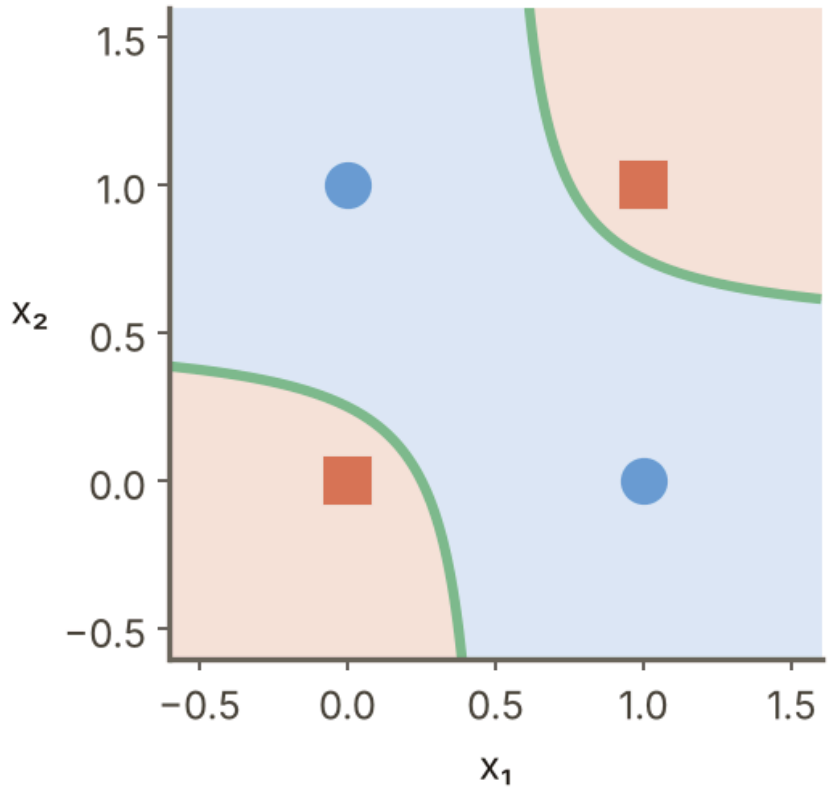
$$g = -¼ + x_1 + x_2 - 2x_1x_2$$

only one new feature, x_1x_2, solves XOR

3. Hyperbolas

$g = 0$ means $x_1 + x_2 - 2x_1x_2 = \frac{1}{4}$, a pair of hyperbola branches. A ($g > 0$) is the band holding $(0, 1)$ and $(1, 0)$; B is the two corners holding $(0, 0)$ and $(1, 1)$. **Example**: at $(0.5, 0.5)$, $g = -0.25 + 1 - 0.5 = +0.25$, on the A side. The same rule is a plane in three dimensions and curved in two.

Figure `lec05-kp15-hyper`: the decision boundary and the two classes.



4. How many terms

For l inputs and order at most r the new coordinates (with the constant) number

$$k = \frac{(l+r)!}{r!\,l!}.$$

Check: $l=2, r=2$: $4!/(2!\,2!) = 6$; $l=3, r=2$: $5!/(2!\,3!) = 10$. For $l=2$, $r = 1..5$ gives 3, 6, 10, 15 and 21 terms.

Figure `lec05-kp15-kcount`: the term count for $l=2$ and $r = 1..5$.

r	k = (2 + r)! / (r! 2!)	terms
1	(3)! / (1! · 2!)	3
2	(4)! / (2! · 2!)	6
3	(5)! / (3! · 2!)	10
4	(6)! / (4! · 2!)	15
5	(7)! / (5! · 2!)	21

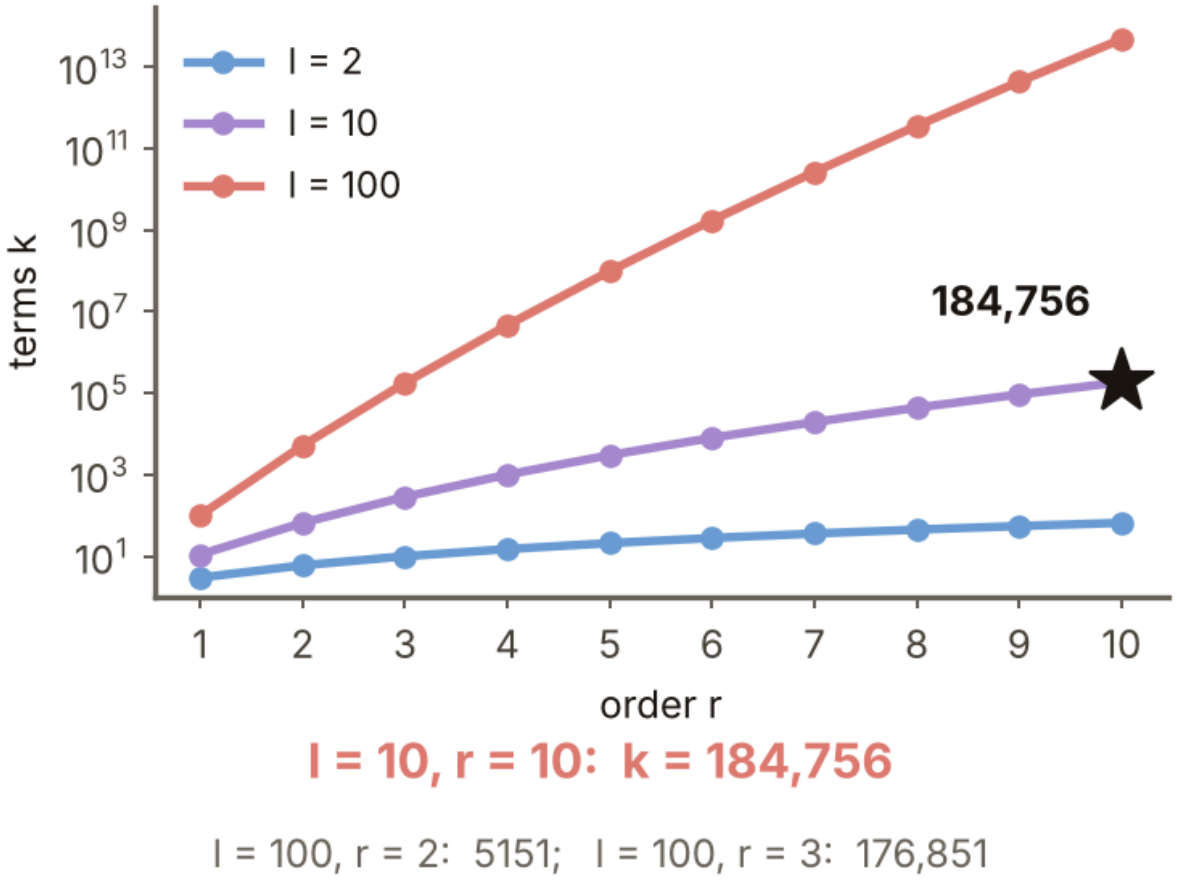
l = 2 inputs; also l = 3, r = 2 gives 10 terms

k counts every product with total power ≤ r

5. Combinatorial explosion

$l = 10, r = 10$ gives $k = 184{,}756$ terms. Others: for $l = 10$, 66 at order 2 and 286 at order 3; for $l = 100$, 5151 and 176,851. It stays small only with few inputs: $l = 2$ gives just 66 terms at $r = 10$. This previews the curse of dimensionality of Chapter 5 (source).

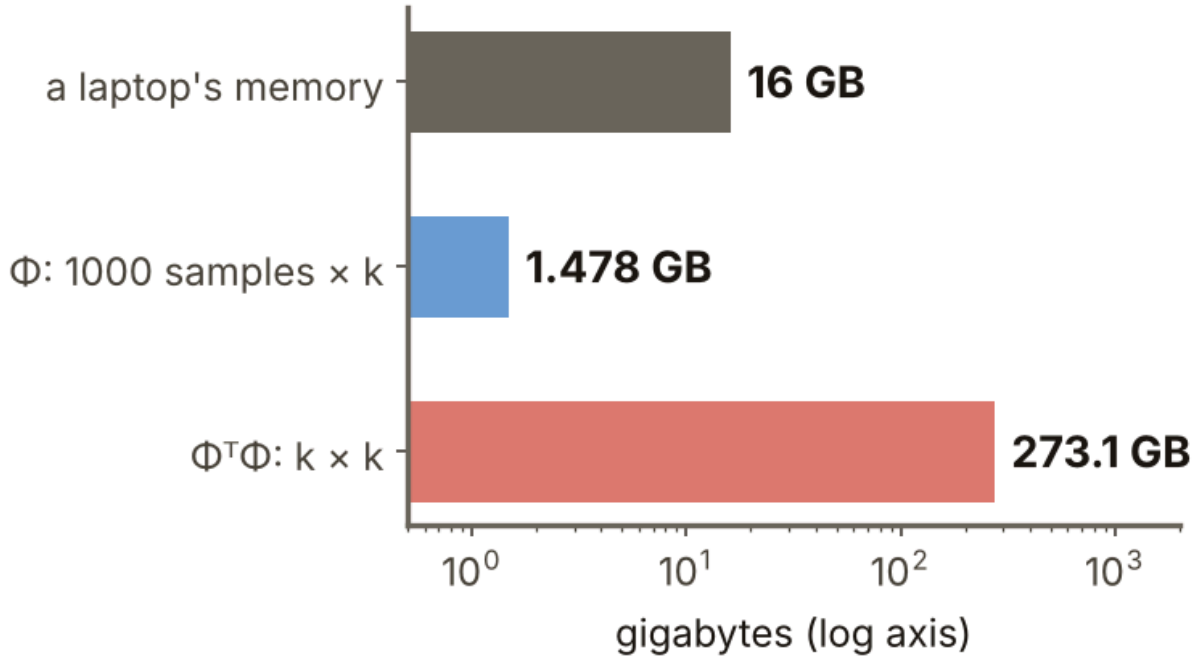
Figure `lec05-kp15-explode`: the term count against the order (log axis).



6. Memory

With $l = 10, r = 10$, $k = 184{,}756$, 8 bytes per number and 1000 samples, the design matrix Φ ($1000 \times k$) takes about 1.48 GB; the $\Phi^T\Phi$ ($k \times k$) of the normal equation from KP13 takes about 273 GB, far more than a typical computer's memory (program).

Figure `lec05-kp15-memory`: memory needs (log axis).



l = 10, r = 10, 8-byte numbers

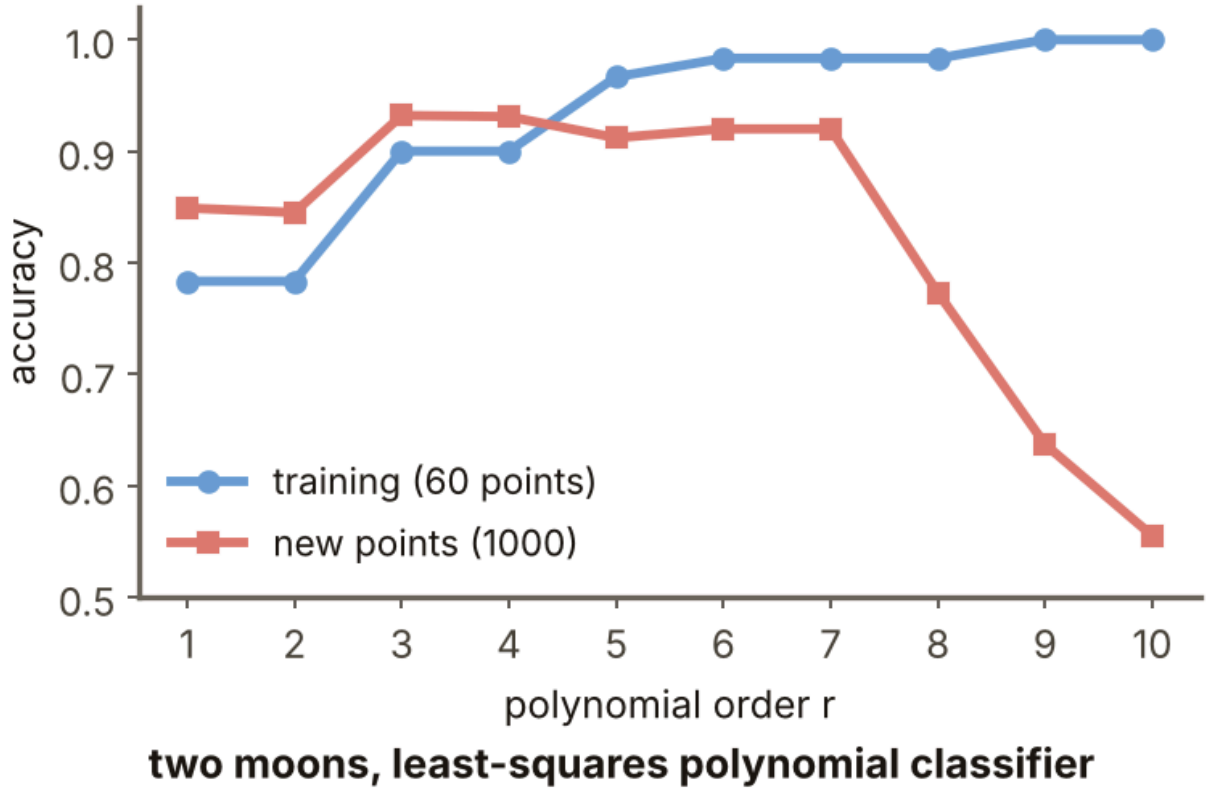
k = 184,756 columns

the normal equation alone needs 273 GB

7. The power

Example (designed here): two moons of 30 points each with noise (sd 0.2), 60 training points; 1000 fresh points; least-squares polynomial classifiers of order $r = 1..10$. Order 1 gets 84.9% on new points and order 3 (10 terms) gets 93.2%: a low-order polynomial already bends around the moons, about 8 points better than a line.

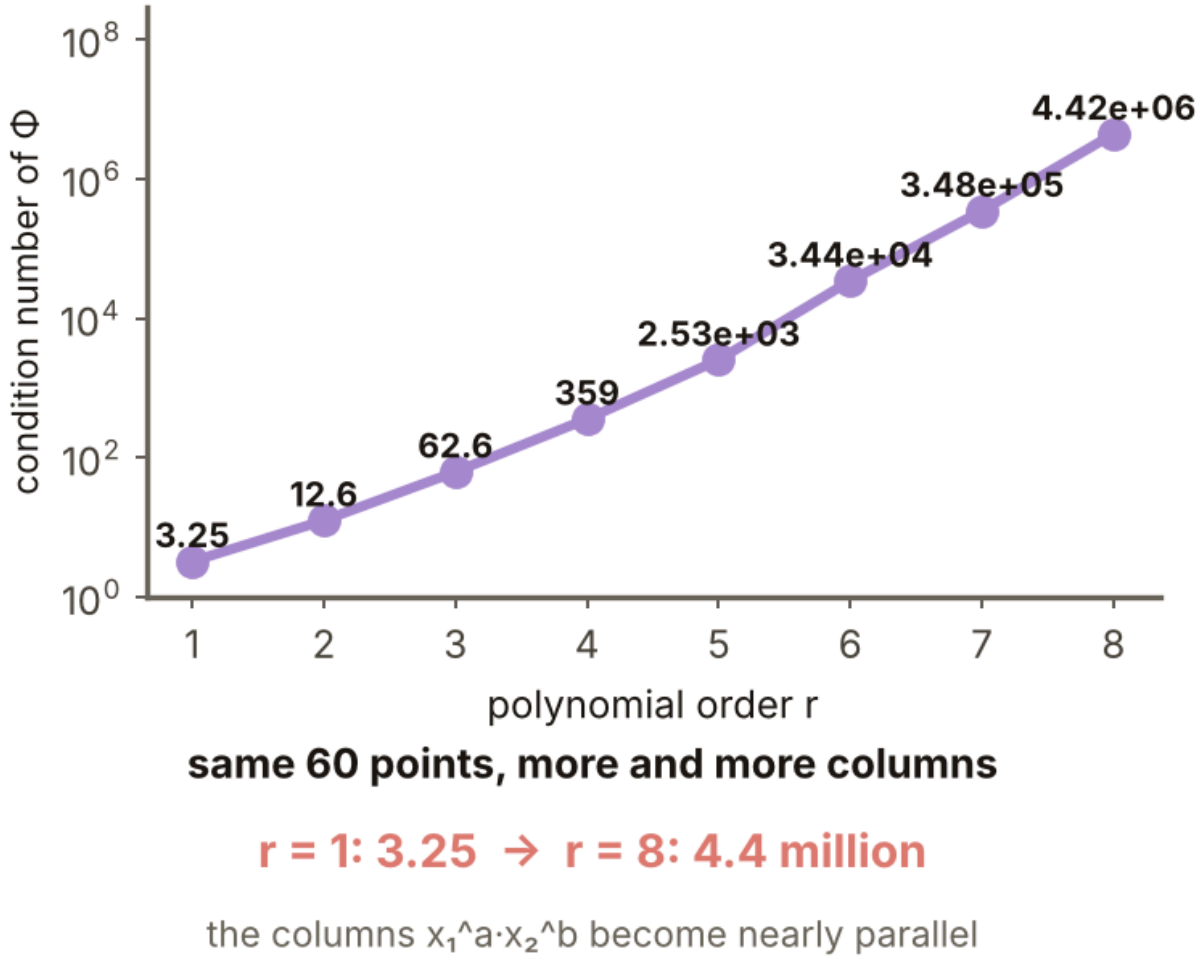
Figure `lec05-kp15-moons`: training and new-point accuracy for orders 1 to 10.



8. Going too far

Condition number: how much a matrix magnifies small errors; the larger, the less stable the solution. On new points $r = 8$ gives 77.3%, $r = 9$ gives 63.7% and $r = 10$ gives 55.5%, while training points are 100% at $r = 9, 10$. **Common misconception**: a higher order is always more accurate. $r = 9$ has 55 terms and $r = 10$ has 66, close to or above the 60 training points, so the model can memorize every point, noise included: overfitting. It is numerically worse too: for the same 60 points the condition number of Φ rises from 3.25 at $r = 1$ to 4.4 million at $r = 8$.

Figure `lec05-kp15-cond`: the condition number of the design matrix against the order (semi-log).



Extension: no new coordinates, practice and worked solution

Kernel preview: $l = 2$, order 2, $x = (1, 2)$, $z = (3, -1)$: $(1 + x \cdot z)^2 = (1 + 3 - 2)^2 = 4$. The other side maps x to $\varphi(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1x_2)$ and term by term $1 + 6 - 4 + 9 + 4 - 12 = 4$, the same. The left side takes one inner product and one power; for $l = 10, r = 10$ the right side is a 184,756-term inner product. Kernel methods (KP19) exist to avoid this explosion.

Figure `lec05-kp15-kernel`: the same number computed two ways.

$$x = (1, 2), \quad z = (3, -1), \quad \text{order } 2$$

$$(1 + x \cdot z)^2 = (1 + 3 - 2)^2 = 4$$

$$\varphi(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1x_2)$$

$$\varphi(x) \cdot \varphi(z) = 1 + 6 - 4 + 9 + 4 - 12 = 4$$

same number, but the left side never builds φ

$l = 10$, $r = 10$:

left: about 10 multiplications and a power

right: a 184,756-term inner product

Practice:

1. For $l = 4$ inputs and order at most $r = 3$, how many terms k?
2. With $g = -\frac{1}{4} + x_1 + x_2 - 2x_1x_2$, which class is the point $(1.5, 1.5)$ ($g > 0$ is A)?

Figure `lec05-kp15-practice`: the setup of the practice.

I = 4 inputs, order r = 3

$$k = (I + r)! / (r! \cdot I!)$$

$$g = -¼ + x_1 + x_2 - 2x_1x_2$$

question 1: how many terms k?

question 2: which class is the point (1.5, 1.5)?

Worked solution of questions 1 and 2: (1) $k = 7!/(3!\,4!) = 5040/(6 \cdot 24) = 35$. (2) $g(1.5, 1.5) = -0.25 + 3 - 4.5 = -1.75 < 0$, class B. (3) Check: 35 is exactly $\binom{7}{3}$ and the point lies in the upper-right B region, the same region as $(1, 1)$.

Figure `lec05-kp15-practice-sol`: the term count and g.

k = 7! / (3! · 4!)	**= 35**
5040 / (6 · 24)	**= 35**
g(1.5, 1.5)	**= −0.25 + 3 − 4.5 = −1.75**
class	**B (g < 0)**

the point is in the upper-right B region, like (1, 1)

Terms and sources

English	中文
Polynomial classifier	多項式分類器
Interaction term	交互項
Condition number	條件數

Source: Chapter 4 KP15, §4.14 (read 2026-10-25). The second-order general form, the term formula $k = (l + r)!/(r!\, l!)$ and 184,756, the XOR rule $g = -\frac{1}{4} + x_1 + x_2 - 2x_1x_2$ with its hyperbolas, the "to remember" line and the pointer to kernel methods follow the source; the order experiment on two moons, the condition numbers, the memory estimate and the kernel check were designed and recomputed for this lecture ([added], (program)).